American Computer Science League

2022-2023 ● Contest 1: Short Problem Solutions ● Elementary Division

1. Computer Number Systems

$1B8_{16} = 8 + 11 * 16 + 1 * 16 * 16 = 8 + 176 + 256 = 440$

D. 440

2. Computer Number Systems

ABE_{16} = 1010 1011 1110 has 8 1s
CDF_{16} = 1100 1101 1111 has 9 1s
7531_8 = 111 101 011 001 has 8 1s
6420_8 = 110 100 010 000 has 4 1s

B. CDF

3. Computer Number Systems

To find how many years it will be, subtract 3746_8 from 5072_8:

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 5072    To subtract 6 from 2, borrow 8 so 10 - 6 = 4 and 6 - 4 = 2.
- 3746    To subtract 7 from 0, borrow 8 so 8 - 7 = 1 and 4 - 3 = 1.
------    Therefore, the answer is 1124.
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B. 1124

4. Computer Number Systems

Converting each number to base 10 in order to compare them is:

a) $1101010_2 = 2 + 8 + 32 + 64 = 106$
b) $5F_{16} = 15 + 5 * 16 = 10615 + 80 = 95$
c) $124_8 = 4 + 2 * 8 + 1 * 64 = 4 + 16 + 64 = 84$
d) 100_{10}

Therefore, the order is 84, 95, 100, 106 or c, b, d, a.

C. c, b, d, a

5. Computer Number Systems

The red components are 40 and 34 and the difference is 16 - 4 = 12 or 0C.
The green components are 9A and 8D and the difference is 26 - 13 = 13 or 0D.
The blue components are F7 and E9 and the difference is 23 - 9 = 14 or 0E.
The smallest difference is 0C.

C. 0C